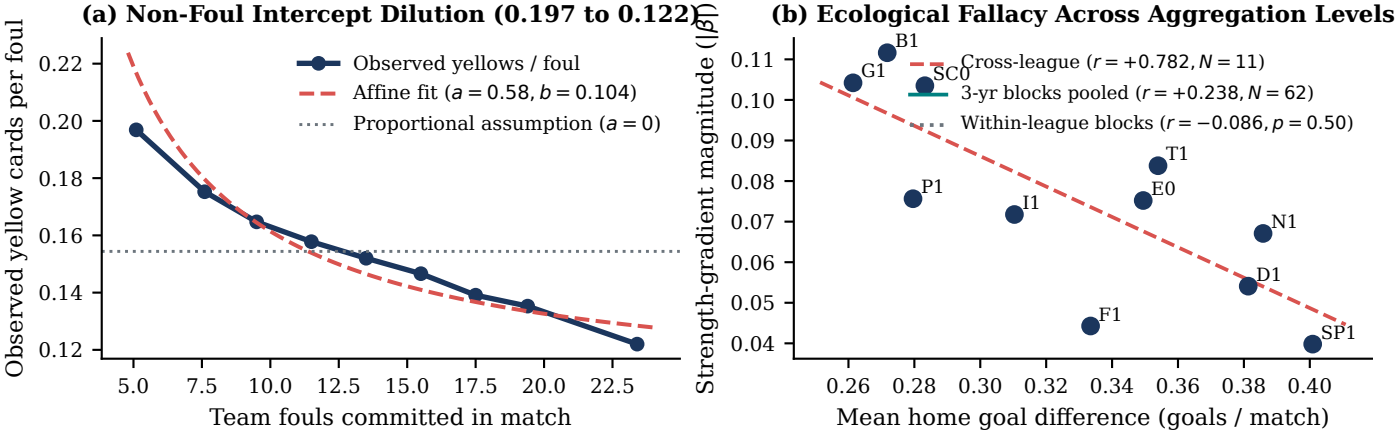
(a) Non-Foul Intercept Dilution (0.197 to 0.122)
Observed yellow cards per foul
Team fouls committed in match
Observed yellows / foul
Affine fit ($a = 0.58, b = 0.104$)
Proportional assumption ($a = 0$)
0.22
0.20
0.18
0.16
0.14
0.12
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
(b) Ecological Fallacy Across Aggregation Levels
Strength-gradient magnitude ($|\beta|$)
Mean home goal difference (goals / match)
Cross-league ($r = +0.782, N = 11$)
3-yr blocks pooled ($r = +0.238, N = 62$)
Within-league blocks ($r = -0.086, p = 0.50$)
B1
G1
SC0
T1
E0
P1
I1
N1
D1
F1
SP1
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.26
0.28
0.30
0.32
0.34
0.36
0.38
0.40